

FIFA WORLD CUP 2026

HOST CITY REVENUE & OPERATIONAL INTELLIGENCE PLATFORM

FICTIONAL CLIENT	FIFA Commercial & Operations Advisory Division
PROJECT TYPE	End-to-End Data & Analytics Consulting Simulation
TARGET ROLES	Data Analyst Business Analyst Data Scientist
TARGET FIRMS	Deloitte KPMG Big 4 D&A Consulting
TOURNAMENTS	2018 Russia 2022 Qatar 2026 USA-Canada-Mexico
DOCUMENT VERSION	1.0 — Project Blueprint & Implementation Guide

PORTFOLIO PROJECT 3 OF 3 | DATA & ANALYTICS CONSULTING TRACK

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Project Overview & Strategic Rationale

What This Project Is

An end-to-end Data & Analytics Consulting simulation that mirrors real Big 4 engagement workflows. The platform benchmarks FIFA World Cup 2026 host city performance against 2018 and 2022 historical baselines, delivering concentration risk scores, revenue projections, and operational load analysis for a fictional FIFA advisory client.

Why This Project Was Chosen

- Most differentiated angle — host city operational analysis is rare on fresher portfolios
- World Cup 2026 is live — temporal relevance makes it a conversation starter through 2027
- Multi-source real data forces genuine data engineering, not synthetic data shortcuts
- Three-tournament longitudinal structure demonstrates analyst maturity over single-snapshot projects
- Domain-agnostic framework — same architecture applies to retail, healthcare, logistics, banking

Portfolio Positioning

- Project 1 — Python ETL + Supabase + Power BI — FinOps / SaaS internal analytics
- Project 2 — Python ML + SQLite + Streamlit — Credit Risk predictive modelling
- Project 3 — Python ETL + BigQuery + Excel + Power BI — Sports operational consulting analytics

No two projects use an identical stack. Each project covers a different problem type, domain, and technical layer — proving versatility across the full data professional skill spectrum.

Project Score

Criteria	Score
Domain Relevance to D&A Consulting	85/100
Data Authenticity & Credibility	82/100
Tool Stack Appropriateness	84/100
Portfolio Differentiation	92/100
Technical Complexity	88/100
Business Narrative Strength	90/100
Interviewer Defensibility	87/100
Completion Probability	80/100
OVERALL ESTIMATED SCORE	87 / 100

Business Problem Statement

The Client Problem

FIFA and host city governments make billion-dollar infrastructure, staffing, and commercial decisions with no unified data view across tournaments. Every World Cup is planned largely from institutional memory and siloed spreadsheets. Post-tournament reports are published 6–12 months after the event — by which time decisions have already been made incorrectly.

Three Core Business Questions This Platform Answers

- Question 1: WHICH HOST CITIES are generating disproportionate revenue dependency — and what is FIFA's structural concentration risk?
- Question 2: IS 2026 TRACKING above or below historical benchmarks on a per-match normalized basis — and where is the variance?
- Question 3: WHERE IS OPERATIONAL LOAD concentrated enough to create delivery risk — and which cities need proactive resource reallocation?

Why Existing Systems Are Insufficient

Capability	Speed	Concentration Risk	Forecasting	Auditability
Manual Number Reading	Hours to days	Impossible at scale	Reactive only	Zero audit trail
This Platform	Real-time	Automated & scored	Prophet projections	Full BigQuery log

Cross-Industry Applicability

The analytical framework is domain-agnostic. The same architecture answers resource allocation and revenue concentration questions in:

- Retail chains — store-level revenue concentration and dependency risk
- Manufacturing / Factories — production scheduling and shift load waveforms
- Hospitals — patient admission surge forecasting by department
- Corporate Real Estate — venue utilization across office campuses
- Supply Chain / Logistics — shipment concentration risk by route or hub
- Banking / Insurance — transaction volume waveforms and claim concentration by geography

The One-Sentence Interview Answer

NOTE: The underlying framework — historical benchmarking, concentration risk scoring, and time-series forecasting — is domain agnostic. I built it on World Cup data because the dataset is rich and the business stakes are real, but the same architecture answers resource allocation and revenue concentration questions in retail, manufacturing, healthcare, or any operation running across multiple locations simultaneously.

Technical Stack & Tool Rationale

Tool / Library	Role in Project
Python (pandas, pytz, geopy, scikit-learn)	Data ingestion, cleaning, schema harmonisation, feature engineering, geographic calculations, timezone co
Facebook Prophet	Time-series revenue forecasting across 3 tournament cycles — purpose-built for sparse seasonal data with
Google BigQuery	Cloud data warehouse — analytical SQL layer with CTEs, window functions, aggregations; pre-calculated v
Excel (Power Query + Pivot)	Revenue estimation model by host city and stage; pivot analysis; deliberately included to demonstrate Big
Power BI (DirectQuery)	Executive dashboard + Operations dashboard — two separate report pages targeting different stakeholder
GitHub Actions	Free automated scheduler — triggers Python scraper post-matchday, pushes updated 2026 data to BigQu
PDF Consulting Report	One-page written narrative with three business recommendations — the layer that separates analyst from c

Why BigQuery Over Supabase (Project 1)

- Deloitte and KPMG D&A; teams run on BigQuery — direct tool alignment with target employer
- Demonstrates stack progression across portfolio — Supabase in Project 1, BigQuery in Project 3
- BigQuery free tier handles this dataset comfortably — 1TB monthly query limit, dataset is ~50,000 rows
- Native DirectQuery connector to Power BI — no intermediary layer needed

Why Prophet Over Linear Regression

Criteria	Linear Regression	Facebook Prophet
Technical Impression	Medium	High
Interview Defensibility	High	Medium
Appropriateness for Time Series	Low	High
Portfolio Differentiation	Low	High
Explainability to Non-Technical	High	Medium
Integration with Power BI	Simple	Moderate

NOTE: Prophet one-liner: 'Revenue patterns across World Cup tournaments follow a 4-year cyclical structure with known seasonal spikes — Prophet is purpose-built for exactly this pattern; linear regression would ignore the temporal dimension entirely.'

Why Two Power BI Report Pages

- Page 1 — EXECUTIVE VIEW: revenue concentration risk, tournament ROI index, three headline KPIs
- Page 2 — OPERATIONS VIEW: match density waveform, city-level staffing load index, logistics pressure score by date
- Different stakeholders need different information from the same data — proving this is a core D&A; consulting skill

Analytical Framework & Methodology

The Normalisation Problem

Direct cross-tournament revenue comparison is methodologically invalid. Russia 2018, Qatar 2022, and USA/Canada/Mexico 2026 represent fundamentally different economic, geographic, cultural, and structural contexts. All comparisons must be normalised before any benchmark is drawn.

Confounding Variables Identified & Controlled

Variable	Treatment
GDP per capita	World Bank annual figures per host nation, vintage-matched to tournament year, converted to constant 2024 USD
Geographic Accessibility	Great circle distance from top 20 footballing nations to host nation — Python geopy; inverted score (higher = more accessible)
Ticket Price to Wage Ratio	Average ticket price divided by host nation average monthly wage — normalises pricing power across economic contexts
Stadium Capacity Utilisation	Actual attendance divided by official FIFA net capacity — consistent definition locked as tickets_sold_official_FIFA_capacity
48-Team Format Adjustment	2026 has 104 matches vs 64 in 2018/2022 — ALL aggregate comparisons normalised to per-match basis
Stage Weighting	Group=1.0x, R32=1.5x, R16=2.0x, QF=3.0x, SF=4.0x, Final=5.0x — captures premium pricing of knockout fixtures
Team Star Power Index	Average FIFA ranking of both teams converted to 0–1 scale — proxy for match commercial demand
Host Nation Match Premium	Binary flag is_host=TRUE for matches involving a host nation team — controls for home crowd attendance premium
Qatar 2022 Structural Outlier	Flagged explicitly: winter tournament, restricted capacity, Stadium 974 temporary — weighted differently in Propher model
Russia 2018 Sanctions Context	Documented as qualitative note — affected certain nations' attendance; not quantified, noted as limitation

Revenue Estimation Model

Actual match-level revenue is never publicly available. A proxy estimation formula is used consistently across all three tournaments:

Estimated Match Revenue = Stadium Capacity x Utilisation Rate x Weighted Average Ticket Price x Stage Weight x Team Star Power Index

NOTE: This formula is documented in the Data Dictionary and referenced explicitly in the PDF Consulting Report. Interviewers respect transparent methodology — hidden assumptions will destroy credibility.

Concentration Risk — HHI Formula

The Herfindahl-Hirschman Index (HHI) is used to score revenue concentration across host cities. This is a standard economic measure used in antitrust regulation and banking supervision.

HHI = sum of squared revenue share of each city
HHI > 0.25 = HIGH CONCENTRATION RISK (standard regulatory threshold)
Example: cities at 30%, 25%, 20% share — HHI = 0.09 + 0.0625 + 0.04 + remaining

Global Primetime Index

Maps every fixture kickoff time against top 10 viewership markets simultaneously. Scoring per market per match:

- 100 points — kickoff falls 7PM–10PM local market time
- 50 points — kickoff falls 5PM–7PM or 10PM–11PM local market time
- 0 points — kickoff falls outside primetime window
- Global Primetime Index = weighted average across all 10 markets, weighted by FIFA broadcast reach figures
- Top 10 markets: Brazil, USA, Germany, France, Argentina, UK, Spain, Mexico, India, China

Prophet Forecasting Layer

- Training data: 2018 (64 matches) and 2022 (64 matches) cumulative tournament-day revenue
- Time index: tournament day number (Day 1 through Day ~32) — not calendar date
- Output: 2026 full-tournament revenue projection with upper and lower confidence intervals
- Validation: Prophet projected vs actual 2026 tracking — forecast error analysis is a finding, not a failure
- Qatar 2022 flagged as outlier in Prophet training — use Prophet built-in outlier weighting parameter

Dataset Inventory & Sources

2026 World Cup — Live Data

Data Need	Source	Format	Access
All 104 fixtures + UTC kickoff	kickoffclock.com	CSV/JSON/Excel	FREE — no login
Live match results (daily)	kaggle.com/datasets/swaptr/fifa-wc-2026-matches	CSVs	FREE — Kaggle login
Relational dataset (full)	kaggle.com/datasets/mominullptr/...	Multi-CSV	FREE — Kaggle login
Live REST + WebSocket API	github.com/rezarahiminia/worldcup2026	API	FREE — token only
Timezone converter all markets	kickoffadventures.com/events/world-cup-26/	Web reference	FREE
Stadium capacities all 16 venues	soccergraph.com 2026 ranked list	Web / manual build	FREE
Pre-tournament rankings Jun 2026	kaggle.com/datasets/lucasyukioimafuko/...	CSV	FREE — filter 2026-06-11

2022 World Cup — Qatar

Data Need	Source	Format	Access
Core match data (64 matches, 59 countries)	kaggle.com/datasets/die9origephit/...	CSV	FREE — zero nulls confirmed
Dedicated attendance per match	kaggle.com/datasets/parasharmanas/...	CSV	FREE — tickets sold definition
Stadium capacity all 8 venues	Manual build — 8 rows	Manual	FREE — 10 minutes
Pre-tournament rankings Oct 2022	kaggle.com/datasets/lucasyukioimafuko/...	CSV	FREE — filter 2022-10-06
Cross-validation layer	datahub.io/football/worldcup	CSV	FREE

2018 World Cup — Russia

Data Need	Source	Format	Access
Comprehensive relational DB	kaggle.com/datasets/joshfjelstul/world-cup-database	Multi-CSV	FREE — AUDIT REQUIRED
Dedicated attendance 1930–2022	kaggle.com/datasets/rajkumarpandey02/...	CSV	FREE — covers 2018 + 2022
Pre-tournament rankings Jun 2018	kaggle.com/datasets/lucasyukioimafuko/...	CSV	FREE — filter 2018-06-01

Supplementary & Economic Data

Data Need	Source	Format	Access
FIFA financial reports (revenue validated)	inside.fifa.com/about-fifa/official-documents/financial-reports	PDFs	FREE
Broadcast viewership 2018	inside.fifa.com/...2018russia/media-releases/...	Web	FREE
Broadcast viewership 2022	fifa.com/fifa-world-cup-qatar-2022-commercial	Web	FREE

Data Need	Source	Format	Access
US host city GDP (metro)	bea.gov/data/gdp/gdp-metropolitan-area	Web	FREE
Mexico city tourism data	datatur.sectur.gob.mx	Web	FREE
World Bank country GDP (USA/CAN/MEX)	data.worldbank.org/v2/country/US;CA;MX/...	JSON	FREE
CPI deflator — constant 2024 USD	databank.worldbank.org	CSV	FREE
2026 ticket prices by stage	theworldcupguide.com/2026-world-cup-ticket-prices/	Web	FREE — phase-1 snapshot
Historical all WC data 1930–2022	kaggle.com/datasets/abecklas/fifa-world-cup	CSV	FREE — Kaggle login

Merge Key Strategy

Primary merge key across all datasets: Match_ID = TournamentYear_Date_HomeTeam_AwayTeam
Example: 2018_0615_RUS_KSA DO NOT rely on dataset-native MatchIDs – they are inconsistent across Kaggle sources. Construct this key manually in Python during the harmonisation step.

Implementation Phases

PHASE 1 | DATASET AUDIT & ENVIRONMENT SETUP | 2–3 Days

Objectives

- Confirm all data sources are downloadable and column-complete before writing code
- Set up isolated Python virtual environment
- Create project folder structure

Actions

- Create Kaggle account — download all datasets listed in Chapter 05
- Open joshfjelstul/world-cup-database — confirm columns: attendance, datetime, stadium, stage, home_team, away_team, city
- Open parasharmanas Qatar 2022 attendance — check null rate in attendance column (flag if above 15%)
- Open mominullptr 2026 dataset — check null rate in attendance column (audit after matchday 3–4 minimum)
- Download FIFA financial reports PDF for 2018 and 2022 — note total ticket revenue and broadcast revenue figures
- Build 8-row Qatar 2022 stadium capacity table manually
- Download BEA US metro GDP data for all 11 US host cities
- Create Python virtual environment: `python -m venv worldcup_env`
- Install all libraries: `pip install pandas prophet scikit-learn pytz geopy google-cloud-bigquery openpyxl reportlab`
- Create requirements.txt immediately after installation

Deliverable

- Audit report — one Excel sheet listing every dataset, column availability (YES/NO), null rate, and data type for each field

PHASE 2 | DATA DICTIONARY & SCHEMA DESIGN | 1–2 Days

Objectives

- Define unified schema before any code is written
- Lock all definitions, assumptions, and data types
- Design BigQuery table structure

Actions

- Build Data Dictionary Excel file — one sheet per tournament
- For every column: source dataset, original column name, unified column name, data type, known issues, imputation method if null
- Lock attendance definition: tickets_sold_official_FIFA across all three tournaments
- Lock revenue definition: estimated_revenue_proxy — document formula explicitly
- Lock capacity definition: FIFA official net capacity — not physical seat count

- Design BigQuery schema: dim_matches, dim_venues, dim_teams, dim_cities, fact_match_revenue, fact_match_operations
- Define all foreign key relationships between tables
- Document 48-team format adjustment: all aggregates to be normalised to per-match basis

Deliverable

- Data Dictionary Excel file — referenced throughout the entire project and included as an appendix in the PDF consulting report

PHASE 3 | PYTHON ETL PIPELINE — INGESTION & CLEANING | 3–4 Days

Objectives

- Ingest all raw data sources into Python
- Clean, harmonise, and merge across three tournaments
- Engineer all calculated features

Actions

- Build schema harmonisation layer first — rename all columns to unified names before any merge
- Handle Cyrillic venue names in 2018 Russia data — transliterate using Python unidecode library
- Convert all monetary figures to constant 2024 USD using World Bank CPI deflator
- Convert all kickoff times to UTC using pytz — never hardcode offsets
- Calculate Geographic Accessibility Score per tournament using geopy great circle distance
- Calculate Team Star Power Index — map FIFA ranking to 0–1 scale (Rank 1 = 1.0, Rank 211 = 0.0)
- Apply Stage Weights: Group=1.0, R32=1.5, R16=2.0, QF=3.0, SF=4.0, Final=5.0
- Calculate Utilisation Rate: attendance / stadium_capacity
- Calculate Estimated Match Revenue using proxy formula
- Build Global Primetime Index using pytz — score each match against 10 markets
- Flag Qatar 2022 structural outlier column: is_qatar_outlier = TRUE
- Flag host nation matches: is_host_nation_match = TRUE
- Build data quality scorecard — completeness score per dataset per column
- Construct Match_ID merge key: TournamentYear_Date_HomeTeam_AwayTeam
- Merge all three tournament datasets into single unified dataframe
- Apply per-match normalisation for all aggregate metrics

Deliverable

- Clean unified CSV — one row per match across all three tournaments — plus data quality scorecard output

PHASE 4 | BIGQUERY — CLOUD WAREHOUSE SETUP | 1–2 Days

Objectives

- Push clean data to BigQuery
- Build analytical SQL layer
- Create pre-aggregated views for Power BI

Actions

- Create BigQuery project — free tier, partition tables by tournament_year to minimise query scan costs
- Push all dimension and fact tables from Python using google-cloud-bigquery library
- Write SQL — CTE calculating HHI concentration risk score per tournament per city
- Write SQL — window function ranking cities by revenue contribution within each tournament
- Write SQL — per-match normalised revenue comparison across all three tournaments
- Write SQL — operational load waveform by tournament day and host city
- Write SQL — Primetime Index aggregation by match, city, and market
- Create BigQuery Views — pre-aggregate all heavy calculations
- Test DirectQuery connection from Power BI to BigQuery before proceeding

Deliverable

- BigQuery dataset with dimension tables, fact tables, analytical views, and confirmed Power BI connection

PHASE 5 | EXCEL REVENUE ESTIMATION MODEL | 1 Day

Objectives

- Build revenue estimation model in Excel
- Demonstrate Big 4 day-to-day tooling
- Produce pivot analysis by city, stage, and nation

Actions

- Import clean unified CSV into Excel via Power Query
- Build revenue estimation model as a structured Excel table with formula columns
- Create named ranges for all input parameters — Stage Weights, Ticket Prices, Capacity figures
- Build pivot table: revenue by host city x tournament year
- Build pivot table: revenue by stage x tournament
- Build pivot table: utilisation rate by venue x tournament
- Create a Summary tab with three headline findings — this feeds directly into the PDF consulting report
- Document all assumptions in a dedicated Assumptions tab

Deliverable

- Excel workbook with Power Query ingestion, revenue model, three pivot tables, and assumptions documentation

PHASE 6 | PROPHET FORECASTING MODEL | 2 Days

Objectives

- Build time-series revenue forecast for 2026 full tournament
- Produce confidence intervals for executive dashboard
- Document forecast methodology for PDF report

Actions

- Aggregate 2018 and 2022 data to cumulative tournament-day level — one row per tournament day
- Prepare Prophet dataframe: ds column (tournament day date), y column (cumulative revenue)
- Flag Qatar 2022 as outlier in Prophet training — use add_seasonality and outlier parameters

- Fit Prophet model on 2018 + 2022 combined training data
- Generate 2026 full-tournament forecast with 80% and 95% confidence intervals
- Compare Prophet projection against actual 2026 cumulative tracking data
- Calculate forecast error and document variance analysis — this is a finding, not a failure
- Push forecast output dataframe to BigQuery forecast table
- Validate: bottom-up per-match estimates should approximately sum to FIFA published total revenue figures

Deliverable

- Prophet model script + forecast output CSV pushed to BigQuery + variance analysis summary

PHASE 7 | POWER BI DASHBOARD — DUAL STAKEHOLDER VIEW | 2–3 Days

Objectives

- Build two report pages targeting different stakeholder types
- Connect to BigQuery via DirectQuery
- Write all DAX measures against confirmed business questions

Page 1 — Executive View

- Revenue Concentration Risk Score (HHI) — gauge visual with HIGH/MEDIUM/LOW threshold bands
- Tournament ROI Index by host city — bar chart normalised per-match
- Prophet Forecast vs Actual overlay — line chart with confidence interval bands
- Top 3 Revenue Cities card visuals with variance vs historical benchmark
- Global Primetime Index heatmap — match x market

Page 2 — Operations View

- Match density waveform by tournament day — area chart showing daily match load
- City-level staffing load index — estimated crowd volume by city by week
- Logistics pressure score by date — composite of match density + expected attendance
- Venue utilisation rate comparison — 2018 vs 2022 vs 2026 side-by-side
- Stage distribution by host city — which cities carry knockout burden

DAX Rules

- Every measure must answer a specific business question — no vanity metrics
- Use CALCULATE + ALL for cross-filter context manipulation in concentration measures
- Use SUMX for row-context revenue aggregation
- Pre-aggregate heavy calculations in BigQuery views — Power BI reads summaries, not raw data

Deliverable

- Published Power BI report — two pages, DirectQuery on BigQuery, shareable URL

PHASE 8 | PDF CONSULTING REPORT & GITHUB README | 1–2 Days

Objectives

- Produce the written insight layer — the single most differentiating element in the portfolio
- Write a consulting narrative, not a technical report
- Frame GitHub README as a consulting case study

PDF Consulting Report Structure

- Section 1 — ENGAGEMENT OVERVIEW: client, problem statement, scope, methodology summary
- Section 2 — DATA LINEAGE: sources, definitions, assumptions, data quality scorecard summary
- Section 3 — KEY FINDINGS: three findings with supporting data — written for a non-technical executive
- Section 4 — BUSINESS RECOMMENDATIONS: exactly three recommendations with commercial justification
- Section 5 — LIMITATIONS: honest documentation of estimation assumptions and data gaps
- Appendix — Data Dictionary reference, formula documentation

GitHub README Rules

- Frame as consulting case study — not technical documentation
- Lead with the business problem, not the tools
- Include architecture diagram showing data flow from source to dashboard
- Document all data sources with direct links
- Include a 'How to Reproduce' section with setup instructions
- Mention tools last — after problem, approach, findings, and recommendations

Deliverable

- One-page PDF consulting report + professional GitHub README — both written for a non-technical audience

Key Analytical Outputs

Output	What It Answers	Where It Appears
Revenue Concentration Risk Score	ROI per tournament — flags which cities drive disproportionate revenue dependency	Executive Page — gauge visual
Per-Match Normalised Benchmark	Cross-tournament revenue comparison adjusted for 48-team format differences	Both pages — bar chart
Prophet Revenue Forecast	2026 full-tournament projection with 80%/95% confidence intervals	Executive Page — line chart
Global Primetime Index	Every fixture scored against 10 broadcast markets — identifies scheduling revenue gaps	Scheduling Page — heatmap
Operational Load Waveform	Match density and crowd volume by city and tournament day	Operations Page — area chart
Venue Utilisation Rate	Attendance as % of FIFA capacity — cross-tournament comparison	Operations Page — comparison bar
Stage-Weighted Revenue Index	Revenue contribution weighted by match importance — identifies top pages	Both pages
Geographic Accessibility Score	Normalisation factor explaining attendance variance across host markets	Methodology documentation
Forecast Error Analysis	Prophet projection vs actual 2026 — variance quantified and explained by causal factors	PDF disclosure report
Data Quality Scorecard	Completeness and confidence score per data source — transparency tool	Dashboards footer / EDC appendix

Interview Defensibility Guide

Every question below will be asked. Prepare these answers before the interview, not during it.

Q: Why football — is this a hobby project?

NOTE: A: Football generates \$40B annually in global commercial activity. The analytical frameworks — concentration risk, longitudinal benchmarking, primetime optimisation — are identical to what I would apply to a retail chain, hospital network, or logistics operation. The domain is sport. The thinking is consulting.

Q: Your data is estimated, not actual — how reliable are your conclusions?

NOTE: A: Every consulting engagement involves estimated data. What matters is transparent methodology. Every estimation is documented with its source, assumption, and confidence level. Conclusions are qualified accordingly — I flag where estimates introduce uncertainty rather than presenting them as facts.

Q: Why didn't you wait for complete 2026 data?

NOTE: A: A real consulting engagement cannot wait for perfect data. The analytical value is in the real-time decision support — seeing whether 2026 is tracking above or below benchmark in week two, not in a post-tournament report six months later. The live element is the point, not a limitation.

Q: How would this scale to a real FIFA deployment?

NOTE: A: Replace the Python scraper with FIFA's internal data API, replace estimated revenue with actual ticketing system feeds, and add role-based access control to the Power BI workspace. The BigQuery schema and DAX measures require zero structural changes — the analytical layer is already production-grade.

Q: What was the hardest technical problem you solved?

NOTE: A: Schema harmonisation across three tournaments with fundamentally different data collection standards. 2018 Russian data uses Cyrillic venue names, 2022 Qatar data has capacity figures in different units, 2026 data is incomplete and being populated live. Building a normalisation pipeline that handles all three consistently without losing data integrity was the core engineering challenge.

Q: What would you do differently with more time?

NOTE: A: Integrate actual broadcaster viewership data via API rather than proxy estimates, and add a Monte Carlo simulation layer to the Prophet forecasting to produce probability distributions rather than point forecasts — giving decision-makers a risk range rather than a single number.

Q: Why Prophet over a simpler model?

NOTE: A: Revenue patterns across World Cup tournaments follow a 4-year cyclical structure with known seasonal spikes. Prophet is purpose-built for exactly this pattern — it handles sparse seasonal data with long gaps natively. Linear regression would ignore the temporal dimension entirely.

Q: Why BigQuery and not a local database?

NOTE: A: The choice signals production intent. BigQuery is what Deloitte and KPMG D&A; teams actually use. A local SQLite database signals a student project. BigQuery signals someone who understands cloud-native data infrastructure.

The Golden Rule For All Interviews

Lead with the business problem. Mention tools last. A recruiter who hears 'I built a Prophet model on BigQuery with Power BI DirectQuery' learns nothing. A recruiter who hears 'I identified that FIFA had no unified system for spotting revenue concentration risk before it became a crisis — so I built one' leans forward.

Missing Data & Resolution Plan

Critical Blockers — Resolve Before Building

Item	Current Status	Resolution Action
2018 column audit	joshfjelstul dataset not yet confirmed	Download and open — confirm attendance, datetime, stadium, stage, city
2026 attendance null rate	Being populated post-match — unknown null %	Audit after matchday 3–4 minimum before building utilisation metric
2022 attendance null rate	Downloaded but not checked	Open parasharmanas dataset — flag if null rate above 15%

Revenue Data Gaps

Gap	Status	Resolution
Match-level actual revenue	Does not exist publicly for any tournament	Use proxy formula — document in Data Dictionary and PDF report
2018 ticket prices	Not yet sourced	Search news archives: 'Russia 2018 World Cup ticket prices category A B
2022 ticket price breakdown	Range confirmed (\$69–\$1,604) but tiers needed	FIFA published ticket category page — cross-reference with Transfermark
Hospitality / VIP revenue	Not public for any tournament	Skip entirely — document as out of scope in PDF report

Broadcast & Viewership Gaps

Gap	Status	Resolution
Match-level viewership	Paywalled everywhere	Use tournament-level totals as market weight proxies only
2022 viewership by market	Not yet confirmed free	Check fifa.com/fifa-world-cup-qatar-2022-commercial for market breakdown
Primetime Index market weight	Not yet calculated	Blocked until broadcast source confirmed

Host City Economic Gaps

Gap	Status	Resolution
Canada host city GDP	Statistics Canada source not identified	Search: 'Statistics Canada GDP metropolitan area Vancouver Toronto 202
Tourism revenue all 16 cities	Partially paywalled	Use hotel occupancy + airport passenger data as proxy — both free
City population all 16 cities	Not yet downloaded	US Census Bureau for US cities, INEGI for Mexico, Statistics Canada for C

Operational Gaps

Gap	Status	Resolution
Geographic Accessibility Score	Formula designed, not yet calculated	Python geopy — one function, apply to tournament dataframe
Weather index per host city	Source not yet identified	Open-Meteo API — free historical weather by coordinates, no login require

Gap	Status	Resolution
2018 Russia stadium capacities	venues not yet confirmed in any dataset	May need manual build — Wikipedia 2018 World Cup stadiums page is rel
2018 Russia city economic data	Sanctions may limit post-2022 sources	World Bank pre-2022 vintage figures available — confirm accessibility

Post-Completion Add-Ons & Enhancements

Complete the core project first. Return to this chapter only after all eight phases are delivered. Every add-on below is genuinely valuable — none of them are filler.

Tier 1 — High Impact, Moderate Effort (Do These First)

Monte Carlo Simulation Layer

Replace Prophet point forecast with probability distribution — run 1,000 simulations with randomised input parameters. Output becomes a revenue risk range rather than a single number. Directly mimics how Deloitte risk advisory teams present financial forecasts to clients.

Automated GitHub Actions Pipeline

Schedule Python scraper via GitHub Actions free tier — triggers post-matchday, pushes 2026 data to BigQuery automatically. Power BI DirectQuery picks up changes on next load. Transforms project from a static build to a live production system.

Sponsor ROI Estimation Layer

Map each match's Global Primetime Index and Team Star Power Index to a Sponsor Exposure Score. Estimates which fixtures delivered maximum brand visibility per sponsorship dollar. Adds a commercial consulting layer that directly maps to KPMG's sports advisory practice.

Tier 2 — High Differentiation, Higher Effort

Sentiment Analysis on Match Commentary

Scrape post-match press coverage or social media sentiment for high-profile fixtures. Correlate sentiment score with attendance variance — does negative pre-match sentiment predict below-capacity attendance? Introduces NLP into the portfolio — the only missing technical domain across all three projects.

Transfer Market Value vs Tournament Performance

Join player transfer value data (Transfermarkt) with national team tournament performance. Analyse whether squad market value predicts attendance uplift for that nation's matches. Adds a player economics layer without building a separate player valuation project.

Carbon Footprint Estimation Layer

Estimate tournament carbon footprint using flight distance data, stadium energy consumption proxies, and visitor volume. Compare 2018 (Russia), 2022 (Qatar air-cooled), 2026 (three-nation travel). Directly maps to ESG consulting — a growing Big 4 service line.

Tier 3 — Portfolio Presentation Enhancements

Notion Case Study Page

Publish the PDF consulting report as a Notion page with embedded Power BI dashboard. Creates a shareable URL that works as a portfolio link in job applications — more accessible than a GitHub repo for non-technical HR reviewers.

Architecture Diagram

Build a visual data flow diagram showing the full pipeline: raw sources → Python ETL → BigQuery → Excel → Power BI → PDF report. Include this in the GitHub README and the Notion page. Architecture diagrams signal systems thinking — a skill Big 4 interviewers specifically probe for.

Video Walkthrough

Record a 3–5 minute screen-capture walkthrough of the Power BI dashboard. Upload to YouTube (unlisted) and link from GitHub README. Most candidates cannot explain their own project coherently under pressure — a recorded walkthrough proves you can.

LinkedIn Article

Write a 600-word LinkedIn article framed as: 'What FIFA's data would tell you if you asked the right questions.' Embed the Notion link. Tag FIFA, Deloitte, KPMG. Visibility drives inbound recruiter interest — especially during an active World Cup.

Priority Order For Add-Ons

- 1. Automated GitHub Actions pipeline — converts project from static to live
- 2. Architecture diagram — immediate, high impact, one afternoon
- 3. Monte Carlo simulation — highest technical credibility boost
- 4. Notion case study page — highest recruiter accessibility
- 5. LinkedIn article — highest visibility during World Cup period
- 6. Sponsor ROI layer — highest D&A; consulting alignment
- 7. Video walkthrough — highest interview preparation value
- 8. Sentiment analysis — highest technical differentiation
- 9. Carbon footprint layer — highest ESG consulting alignment
- 10. Transfer market value layer — optional, only if time permits